Guangdong Technion
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Homework 10 - Deadline January 10

1. For which positive integers k is the series $\sum_{n=1}^{\infty} \frac{(n!)^2}{(kn)!}$ convergent?

2. Use any test to determine whether the series is absolutely convergent, conditionally convergent, or divergent:

a) $\sum_{n=1}^{\infty} \frac{(-1)^n}{n \ln(n+1)}$

b) $\sum_{n=1}^{\infty} \left(\frac{\ln n}{n}\right)^3$

c) $\sum_{n=1}^{\infty} \frac{e^{2n}}{6^{n-1}}$

d) $\sum_{n=1}^{\infty} \ln\left(\frac{n}{n+1}\right)$

3. Determine if the following series converge or diverge

(a) $\sum_{n=1}^{\infty} a_n$, where $a_1 = \frac{1}{2}$ and for all $n \in \mathbb{N}$, $a_{n+1} = \left(\frac{n^2+4}{n^2+3}\right)^{n^2/2} a_n$.

(b) $\sum_{n=1}^{\infty} (-1)^{n-1} b_n$, where $b_n = \frac{1}{n}$ if n is odd and $b_n = \frac{1}{n^2}$ if n is even.
Why does the Alternating Series Test not apply?

4. Find the radius of convergence and the interval of convergence of each of the following power series:

a) $\sum_{n=1}^{\infty} \frac{(-1)^n}{(3n^2+1)} x^n$

b) $\sum_{n=1}^{\infty} \frac{(-1)^n}{(3n+1)} (3x+1)^n$.

5. Find the Maclaurin series and its interval of convergence for the functions

a) $f(x) = \frac{x^2}{(1-2x)}$

b) $F(x) = \int_0^x \left[e^{-3t^2} + \frac{4}{4+t^2}\right] dt$.

6. Use the Taylor series of $f(x) = e^x$ to find an approximation of $\sqrt{e}$ with an accuracy of 10^{-5}.